了解 SnapDrive for UNIX 守□□程

Snapdrive for Unix

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了解 SnapDrive for UNIX 守□□程

在□行任何 SnapDrive for UNIX 命令之前，□必□了解 Web 服□和守□□程以及如何使用它□。所有 SnapDrive for UNIX 命令均使用守□□程服□□行。在 AIX 主机上使用 SnapDrive for UNIX 之前，必□□□守□□程，以便 SnapDrive for UNIX 能□与其他 NetApp 和非 NetApp □品无□安全地集成。

什□是 Web 服□和守□□程

SnapDrive for UNIX Web 服□□所有 NetApp SnapManager 和第三方□品提供了一个□一的界面，可与 SnapDrive for UNIX 无□集成。要在□用于 UNIX 的 SnapDrive 中使用命令行界面（ CLI ）命令，□需要□□守□□程。

各□ NetApp SnapManager □品都使用命令行界面（ CLI ）与 SnapDrive for UNIX □行通信。使用命令行界面会限制 SnapManager 和 SnapDrive for UNIX 的性能和易管理性。使用 SnapDrive for UNIX 守□□程□，所有命令都是一个唯一的□程。守□□程服□不会影□ SnapDrive for UNIX 命令的使用方式。

SnapDrive for UNIX Web 服□允□第三方□用程序与 SnapDrive for UNIX 无□集成。它□使用 API 与 SnapDrive for UNIX □行交互。

□□守□□程□， SnapDrive for UNIX 守□□程会首先□□□守□□程是否正在□行。如果守□□程未□行，□□□守□□程。如果守□□程已在□行，而□□□□□它，□ SnapDrive for UNIX 将□示以下消息：

SnapDrive 守□□程已在□行`

□可以□□守□□程的状□以□看 SnapDrive for UNIX 是否正在□行。在决定□□守□□程之前，□先□□状□。如果非 root 用□□□□□状□，□ SnapDrive for UNIX 将□□□用□的凭据并□示消息：

只有 root 用□` 才能□看 SnapDrive 守□□程状□

当□□□停止守□□程□， SnapDrive for UNIX 会□□□的凭据。如果□不是 root 用□，□ SnapDrive for UNIX 将□示此消息

SnapDrive 守□□程只能由 root 用□` 停止

停止守□□程后，必□重新□□ SnapDrive for UNIX 守□□程，□配置文件或任何模□所做的任何更改才能生效。如果非 root 用□□□重新□□ SnapDrive for UNIX 守□□程，□ SnapDrive for UNIX 将□□□用□的凭据并□示消息

SnapDrive 守□□程只能由 root 用□` 重新□□

正在□□守□□程的状□

□可以□□守□□程的状□以□看守□□程是否正在□行。如果守□□程已在□行，□在更新 SnapDrive for UNIX 配置文件之前，无需重新□□它。

□必□以 root 用□身□登□。

□□

1. □□守□□程的状□：

   ` * 快照状□ *`

□□ SnapDrive for UNIX 守□□程

□必□先□□并□行 SnapDrive for UNIX 守□□程，然后才能使用任何 SnapDrive for UNIX 命令。

□必□以 root 用□身□登□。

□□

1. □□守□□程：

   ` * 快照□□ *`

更改默□守□□程密□

SnapDrive for UNIX 会分配一个默□守□□程密□，□可以□后更改□密□。此密□存□在一个加密文件中，其中只会□ root 用□分配□取和写入□限。更改密□后，必□手□通知所有客□端□用程序。

□必□以 root 用□身□登□。

□□

1. 更改默□密□：

   ` * 快照密□ *`

2. □入密□。
3. □□密□。

正在停止守□□程

如果更改 SnapDrive for UNIX 配置文件，□必□停止并重新□□守□□程。□可以非□制或□制停止守□□程。

不□制停止守□□程

如果 SnapDrive for UNIX 配置文件□生更改，□必□停止守□□程，配置文件更改才能生效。停止并重新□□守□□程后，配置文件中的更改将生效。如果不□制停止守□□程，□所有排□的命令都可以完成□行。收到停止□求后，不会□行任何新命令。

□必□以 root 用□身□登□。

1. □入以下命令以非□制停止守□□程：

`* 快照停止 *`

□制停止守□□程

如果□不想等待所有命令完成□行，□可以□制停止守□□程。收到□制停止守□□程的□求后， SnapDrive for UNIX 守□□程将取消正在□行或正在排□的所有命令。□制停止守□□程□，系□的状□可能未定□。不建□使用此方法。

□必□以 root 用□身□登□。

□□

1. □制停止守□□程：

   `* snapdrived -force stop*`

正在重新□□守□□程

□必□在停止守□□程后重新□□它，以便□配置文件或其他模□所做的更改生效。只有在完成正在□行且□于□列中的所有命令后， SnapDrive for UNIX 守□□程才会重新□□。收到重新□□□求后，不会□行任何新命令。

- □保以 root 用□身□登□。
- □保同一主机上未并行□行任何其他会□。在□□情况下， `snapdrived restart` 命令将挂起系□。

□□

1. □入以下命令重新□□守□□程：

   `* 快照重新□□ *`

□制守□□程重新□□

□可以□制此守□□程重新□□。□制重新□□守□□程将停止□行所有正在□行的命令。

□保以 root 用□身□登□。

□□

1. □入以下命令□制重新□□守□□程：

   `* snapdrived -force restart*`

   收到□制重新□□□求后，守□□程将停止所有正在□行和正在排□的命令。只有在取消□行所有正在□行的命令后，守□□程才会重新□□。

使用 HTTPS 安全守□□程通信

□可以使用 HTTPS □行安全的 Web 服□和守□□程通信。通□在 `snapdrive.conf` 文件中□置一些配置□量，并生成和安装自□名□□或 CA □名□□，可以□用安全通信。

□必□在 `snapdrive.conf` 文件中指定的路径□提供自□名或 CA □名□□。要使用 HTTPS □行通信，必□在 `snapdrive.conf` 文件中□置以下参数：

- `use-https-to-sdU-daemon=on`
- `contact-https-port-sdU-daemon = 4095`
- `sdu-daemon-certificate-path=/opt/NetApp/SnapDrive/SnapDrive.pem`

> SnapDrive 5.0 for UNIX 及更高版本支持 HTTPS 用于守□□程通信。默□情况下，此□□□置□ `off` 。

生成自□名□□

SnapDrive for UNIX 守□□程服□要求□生成用于身□□□的自□名□□。与命令行界面通信□需要□行此身□□□。

□□

1. 生成 RSA 密□：

   ` * $ openssl genrsa 1024 > host.key $ chmod 400 host.key*``

   ```
   # openssl genrsa 1024 > host.key Generating
   RSA private key, 1024 bit long modulus
   ..............................++++++ ...++++++ e is 65537(0x10001)
   # chmod 400 host.key
   ```

2. □建□□：

   ` * $ openssl req -new -x509 -nodes -SHA1 -days 365 -key host.key > host.cert*`

   使用 ` -new` ， ` -x509` 和 ` -nodes` □□□建未加密□□。` - 天` □□指定□□保持有效的天数。

3. 当系□要求□填写□□的 x509 数据□，□□入□的本地数据：

```
# openssl req -new -x509 -nodes -sha1 -days 365 -key host.key >
host.cert
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a
DN. There are quite a few fields
but you can leave some blank For some fields there will be a default
value, If you enter '.', the field will be left blank.

Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:California
Locality Name (eg, city) []:Sunnyvale
Organization Name (eg, company) [Internet Widgits Pty Ltd]:abc.com
Organizational Unit Name (eg, section) []:
Common Name (eg, YOUR name) []:localhost
Email Address []:postmaster@example.org
```

ⓘ `Common Name` □必□□ *localhost* 。

4. 提取元数据（可□）。

```
$ openssl x509 -noout -fingerprint -text < host.cert > host.info
```

□可以保存□□元数据以供日后快速参考。

5. 将密□和□□数据□合使用。

SnapDrive for UNIX 要求密□和□□数据位于同一文件中。□合文件必□作□密□文件□行保□。

` * $ cat host.cert host.key > host.pem \*`

` * 和 rm host.key*`

` * $ chmod 400 host.pem*`

```
# cat host.cert host.key >  /opt/NetApp/snapdrive.pem
# rm host.key rm: remove regular file `host.key'? y
# chmod 400 /opt/NetApp/snapdrive.pem
```

6. 将守□□程□□的完整路径添加到 `snapdrive.conf` 文件的 ` *sdU-daemon-certificate-path*` □量。

生成 CA □名的□□

SnapDrive for UNIX 守□□程服□要求□生成 CA □名的□□，才能成功□行守□□程通信。□

必□在 `snapdrive.conf` 文件中指定的路径□提供 CA □名□□。

- □必□以 root 用□身□登□。
- □必□已在 `snapdrive.conf` 文件中□置以下参数，才能使用 HTTPS □行通信：
  - use-https-to-sdU-daemon=on
  - contact-https-port-sdU-daemon = 4095
  - sdU-daemon-certificate-path=` /opt/NetApp/SnapDrive/SnapDrive.pem`

□□

1. 以 pem 格式生成新的未加密 RSA 私□：

   ` * $ openssl genrsa -out privkey.pem 1024*`

   ```
   Generating RSA private key, 1024 bit long modulus
    ....................++++++ ....................................++++++
   e is 65537 (0x10001)
   ```

2. 配置 ` /etc/ssl/openssl.cnf` 以□建 CA 私□和□□ vi ` /etc/ssl/openssl.cnf` 。
3. 使用 RSA 私□□建未□名的□□：

   ` * $ openssl req -new -x509 -key privkey.pem -out cert.pem*`

   ```
   You are about to be asked to enter information that will be
   incorporated into your certificate request.
   What you are about to enter is what is called a Distinguished Name or
   a DN.
   There are quite a few fields but you can leave some blank For some
   fields there will be a default value, If you enter '.', the field
   will be left blank.
   -----
   Country Name (2 letter code) [XX]:NY
   State or Province Name (full name) []:Nebraska Locality Name (eg,
   city) [Default City]:Omaha Organization Name (eg, company) [Default
   Company Ltd]:abc.com Organizational Unit Name (eg, section) []:
   Common Name (eg, your name or your server's hostname) []:localhost
   Email Address []:abc@example.org
   ```

4. 使用私□和□□□建 CSR ：

   ` * cat cert.pem privkey.pem | openssl x509 -x509 tor均衡 器 -signkey privkey.pem -out certreq.csd*`

   ```
   Getting request Private Key Generating certificate request
   ```

5. 使用□□□建的 CSR 使用 CA □用密□□□□□行□名：

` * $ openssl ca -in certreq.csr -out newcert.pem*`

```
Using configuration from /etc/pki/tls/openssl.cnf Check that the
request matches the signature Signature ok Certificate Details:
         Serial Number: 4096 (0x1000)
         Validity
            Not Before: May 17 06:02:51 2015 GMT
             Not After : May 16 06:02:51 2016 GMT
             Subject:
             countryName               = NY
             stateOrProvinceName       = Nebraska
             organizationName          = abc.com
             commonName                = localhost
             emailAddress              = abc@example.org
             X509v3 extensions:
             X509v3 Basic Constraints:
                 CA:FALSE
             X509v3 Key Usage:
                 Digital Signature, Non Repudiation, Key Encipherment
             Netscape Comment:
                 OpenSSL Generated Certificate
             X509v3 Subject Key Identifier:

FB:B0:F6:A0:9B:F2:C2:BC:50:BF:45:B2:9D:DB:AA:3B:C5:07:5B:7F
             X509v3 Authority Key Identifier:

 keyid:FB:B0:F6:A0:9B:F2:C2:BC:50:BF:45:B2:9D:DB:AA:3B:C5:07:5B:7F

 Certificate is to be certified until May 16 06:02:51 2016 GMT (365
 days) Sign the certificate? [y/n]:y


 1 out of 1 certificate requests certified, commit? [y/n]y Write out
 database with 1 new entries Data Base Updated
```

6. 安装 SSL 服□器要使用的□名□□和□用密□。

```
The newcert.pem is the certificate signed by your local CA that you can
then use in an
ssl server:
( openssl x509 -in newcert.pem; cat privkey.pem ) > server.pem
ln -s server.pem `openssl x509 -hash -noout -in server.pem`.0 # dot-zero
( server.pem refers to location of https server certificate)
```

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